

Customer Transaction & Disbursement Insights Summary

Senior Data Analyst Case Submission

02/05/2025

Summary & Tools

- Key types of data analyzed (transactions, customer info)
- Tools and techniques used
 - Matplotlib & Seaborn for visualizations
 - Python 3.9+ for ETL scripting and analysis
 - PostgreSQL as the target data warehouse
 - psycopg2 for PostgreSQL integration
 - pandas for data cleaning and transformation
 - Microsoft Excel

Folder structure for easy navigation

```
Senior_Data_Analyst_ETL_Case_Submission/  
├── README.md  
├── ETL_Process_Design.pdf  
├── data/  
├── images/  
├── scripts/  
├── sql/  
├── output/  
└── raport_summary.pdf
```

PART 1: ETL PRROCESS DEDIGN

Refer to ETL_Process_Design.pdf for a detailed explanation.

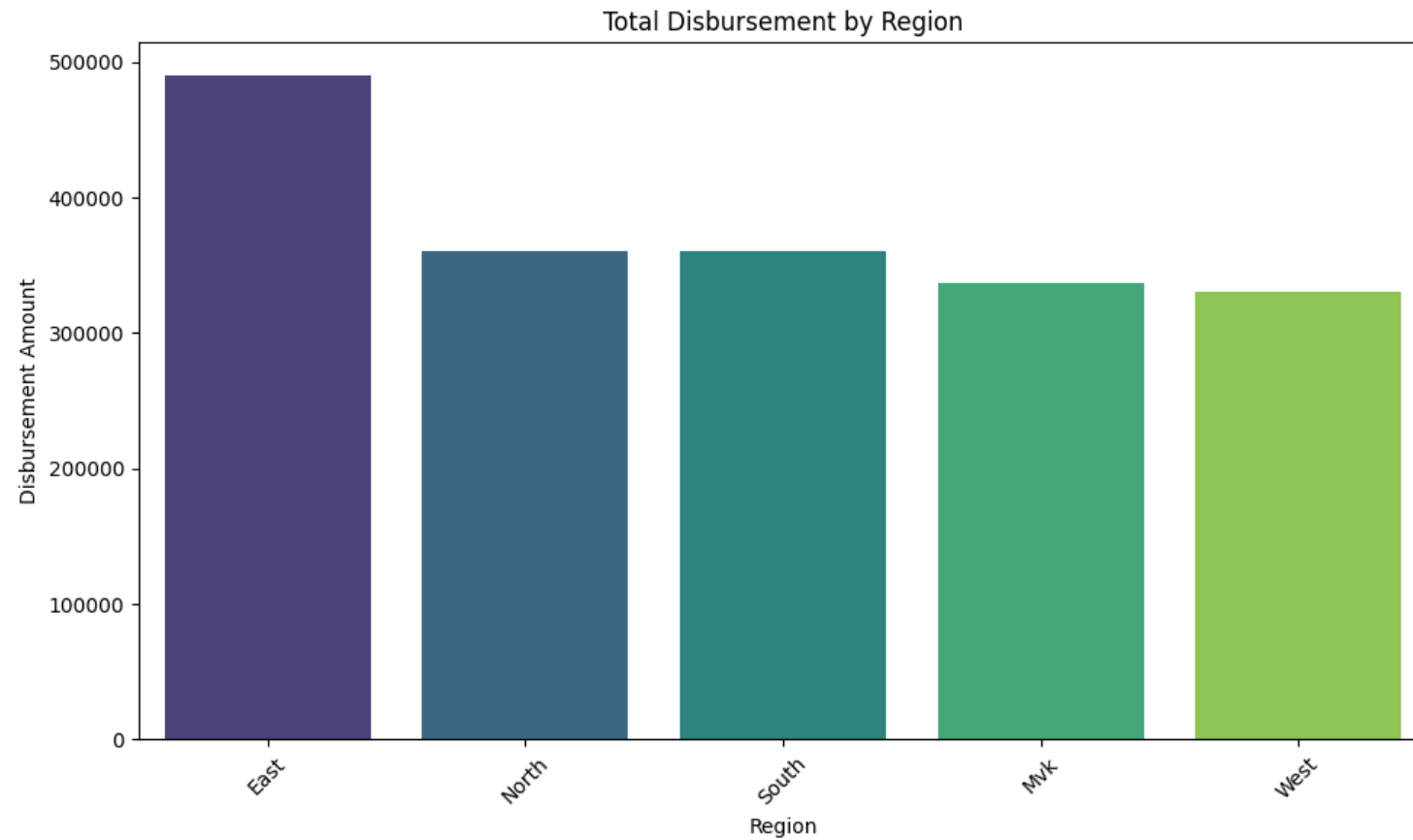
PART 2: SCRIPTING FOR DATA ANALYSIS

The scripts for loading the dataset, calculating disbursement allocations per customer, and other related tasks are stored in the scripts folder. The resulting output files are saved in the output folder

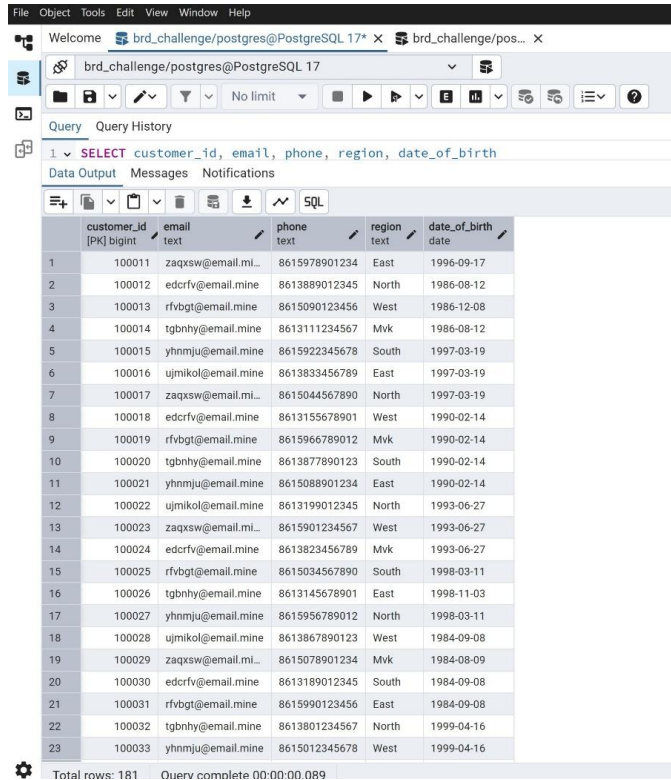
PART 3: DATA ANALYSIS WITH DB QUERYING

The queries are stored in the sql folder, and the output results are saved in the output folder.

Disbursement by Region



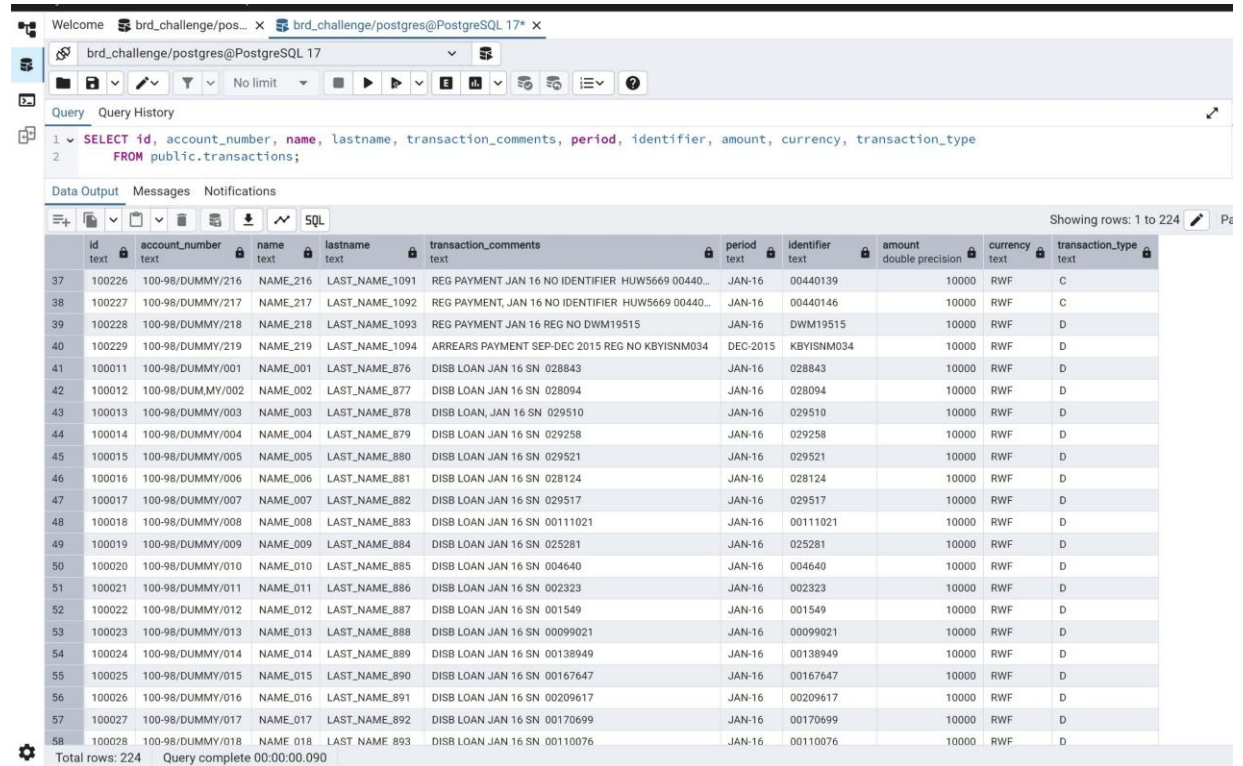
Data warehouse tables (Postgres)



The screenshot shows a database client interface with a query window and a results pane. The query is a SELECT statement that retrieves customer information. The results pane displays a table with 23 rows of data. The columns are: customer_id, email, phone, region, and date_of_birth. The data is sorted by customer_id in ascending order. The status bar at the bottom indicates that there are 181 total rows and the query is complete.

	customer_id [PK] bigint	email text	phone text	region text	date_of_birth date
1	100011	zaqsw@email.mi...	8615978901234	East	1996-09-17
2	100012	edcrfv@email.mine	8613889012345	North	1986-08-12
3	100013	rfvbg@email.mine	8615090123456	West	1986-12-08
4	100014	tgbnhy@email.mine	8613111234567	Mvk	1986-08-12
5	100015	yhnrmju@email.mine	8615922345678	South	1997-03-19
6	100016	ujmikol@email.mine	8613833456789	East	1997-03-19
7	100017	zaqsw@email.mi...	8615044567890	North	1997-03-19
8	100018	edcrfv@email.mine	8613155678901	West	1990-02-14
9	100019	rfvbg@email.mine	8615966789012	Mvk	1990-02-14
10	100020	tgbnhy@email.mine	8613877890123	South	1990-02-14
11	100021	yhnrmju@email.mine	8615088901234	East	1990-02-14
12	100022	ujmikol@email.mine	8613199012345	North	1993-06-27
13	100023	zaqsw@email.mi...	8615901234567	West	1993-06-27
14	100024	edcrfv@email.mine	8613823456789	Mvk	1993-06-27
15	100025	rfvbg@email.mine	8615034567890	South	1998-03-11
16	100026	tgbnhy@email.mine	8613145678901	East	1998-11-03
17	100027	yhnrmju@email.mine	8615956789012	North	1998-03-11
18	100028	ujmikol@email.mine	8613867890123	West	1984-09-08
19	100029	zaqsw@email.mi...	8615078901234	Mvk	1984-08-09
20	100030	edcrfv@email.mine	8613189012345	South	1984-09-08
21	100031	rfvbg@email.mine	8615990123456	East	1984-09-08
22	100032	tgbnhy@email.mine	8613801234567	North	1999-04-16
23	100033	yhnrmju@email.mine	8615012345678	West	1999-04-16

Total rows: 181 Query complete 00:00:00.089



The screenshot shows a database client interface with a query window and a results pane. The query is a SELECT statement that retrieves transaction information. The results pane displays a table with 58 rows of data. The columns are: id, account_number, name, lastname, transaction_comments, period, identifier, amount, currency, and transaction_type. The data is sorted by id in ascending order. The status bar at the bottom indicates that there are 224 total rows and the query is complete.

	id text	account_number text	name text	lastname text	transaction_comments text	period text	identifier text	amount double precision	currency text	transaction_type text
37	100226	100-98/DUMMY/216	NAME_216	LAST_NAME_1091	REG PAYMENT JAN 16 NO IDENTIFIER HUW5669 00440...	JAN-16	00440139	10000	RWF	C
38	100227	100-98/DUMMY/217	NAME_217	LAST_NAME_1092	REG PAYMENT, JAN 16 NO IDENTIFIER HUW5669 00440...	JAN-16	00440146	10000	RWF	C
39	100228	100-98/DUMMY/218	NAME_218	LAST_NAME_1093	REG PAYMENT JAN 16 REG NO DWM19515	JAN-16	DWM19515	10000	RWF	D
40	100229	100-98/DUMMY/219	NAME_219	LAST_NAME_1094	ARREARS PAYMENT SEP-DEC 2015 REG NO KBYISNM034	DEC-2015	KBYISNM034	10000	RWF	D
41	100011	100-98/DUMMY/001	NAME_001	LAST_NAME_876	DISB LOAN JAN 16 SN 028843	JAN-16	028843	10000	RWF	D
42	100012	100-98/DUM,MY/002	NAME_002	LAST_NAME_877	DISB LOAN JAN 16 SN 028094	JAN-16	028094	10000	RWF	D
43	100013	100-98/DUMMY/003	NAME_003	LAST_NAME_878	DISB LOAN, JAN 16 SN 029510	JAN-16	029510	10000	RWF	D
44	100014	100-98/DUMMY/004	NAME_004	LAST_NAME_879	DISB LOAN JAN 16 SN 029258	JAN-16	029258	10000	RWF	D
45	100015	100-98/DUMMY/005	NAME_005	LAST_NAME_880	DISB LOAN JAN 16 SN 029521	JAN-16	029521	10000	RWF	D
46	100016	100-98/DUMMY/006	NAME_006	LAST_NAME_881	DISB LOAN JAN 16 SN 028124	JAN-16	028124	10000	RWF	D
47	100017	100-98/DUMMY/007	NAME_007	LAST_NAME_882	DISB LOAN JAN 16 SN 029517	JAN-16	029517	10000	RWF	D
48	100018	100-98/DUMMY/008	NAME_008	LAST_NAME_883	DISB LOAN JAN 16 SN 00111021	JAN-16	00111021	10000	RWF	D
49	100019	100-98/DUMMY/009	NAME_009	LAST_NAME_884	DISB LOAN JAN 16 SN 025281	JAN-16	025281	10000	RWF	D
50	100020	100-98/DUMMY/010	NAME_010	LAST_NAME_885	DISB LOAN JAN 16 SN 004640	JAN-16	004640	10000	RWF	D
51	100021	100-98/DUMMY/011	NAME_011	LAST_NAME_886	DISB LOAN JAN 16 SN 002323	JAN-16	002323	10000	RWF	D
52	100022	100-98/DUMMY/012	NAME_012	LAST_NAME_887	DISB LOAN JAN 16 SN 001549	JAN-16	001549	10000	RWF	D
53	100023	100-98/DUMMY/013	NAME_013	LAST_NAME_888	DISB LOAN JAN 16 SN 00099021	JAN-16	00099021	10000	RWF	D
54	100024	100-98/DUMMY/014	NAME_014	LAST_NAME_889	DISB LOAN JAN 16 SN 00138949	JAN-16	00138949	10000	RWF	D
55	100025	100-98/DUMMY/015	NAME_015	LAST_NAME_890	DISB LOAN JAN 16 SN 00167647	JAN-16	00167647	10000	RWF	D
56	100026	100-98/DUMMY/016	NAME_016	LAST_NAME_891	DISB LOAN JAN 16 SN 00209617	JAN-16	00209617	10000	RWF	D
57	100027	100-98/DUMMY/017	NAME_017	LAST_NAME_892	DISB LOAN JAN 16 SN 00170699	JAN-16	00170699	10000	RWF	D
58	100028	100-98/DUMMY/018	NAME_018	LAST_NAME_893	DISB LOAN JAN 16 SN 00110076	JAN-16	00110076	10000	RWF	D

Total rows: 224 Query complete 00:00:00.090

Other Key Findings

A. Disbursement Trends

- Total disbursement volume across all regions: **2,217,030 RWF**
- Highest disbursing region: **East Region**
- Peak periods: **JAN-16**

B. Customer Behavior

- Average transaction amount: **9,897.46 RWF**
- Consistent high-volume customers: **Customer_ID 100156, 100215**

C. Regional Patterns

- Region **East** shows higher average disbursements but fewer transactions
- Region **Mvk** has more customers but lower disbursement per capital